Introduction to JavaScript

1. **What is JavaScript?**
    a. JavaScript is a high-level scripting as well as programming language which is used to create interactive web pages.
    b. JavaScript is an object-based scripting language which is lightweight and cross-platform.
    c. With the help of JS we are able to add functionality and behaviour to HTML and CSS webpages.
    d. It is the only language which is understood by the browser.
    e. JavaScript is also known as the scripting language.
    f. It is the language you can use at the browser side as well as the server side.
    g. It is the most common and popular language used right now.
    h. A lot of frameworks and libraries are based on javascript. It can be used for both frontend and backend.
        i. Frontend → React.js, Angular.js, Next.js, etc.
        ii. Backend → Node.js, Express.js, etc.

2. **History of JS:**
    a. In the early days of the web, HTML and CSS were used to create static web pages.
    b. HTML provided the structure, and CSS controlled the presentation.
    c. There was no way to create interactive or dynamic behavior directly in the browser.
    d. Users had to reload entire pages to see updates or interact with content, which was a significant limitation.

e. Creation of JavaScript :

i. 1993 ⇒ *Mosaic* , the first popular web browser with a graphical user interface, was released. It made the web accessible to non-technical people and spurred the rapid growth of the World Wide Web.

ii. 1994 ⇒ The lead developers of Mosaic founded *Netscape Corporation* and released *Netscape Navigator* , a more polished browser.

iii. 1995 ⇒ Netscape decided to add a programming language to Navigator.

iv. Two approaches were pursued:

v. Collaborating with Sun Microsystems to embed the Java language.

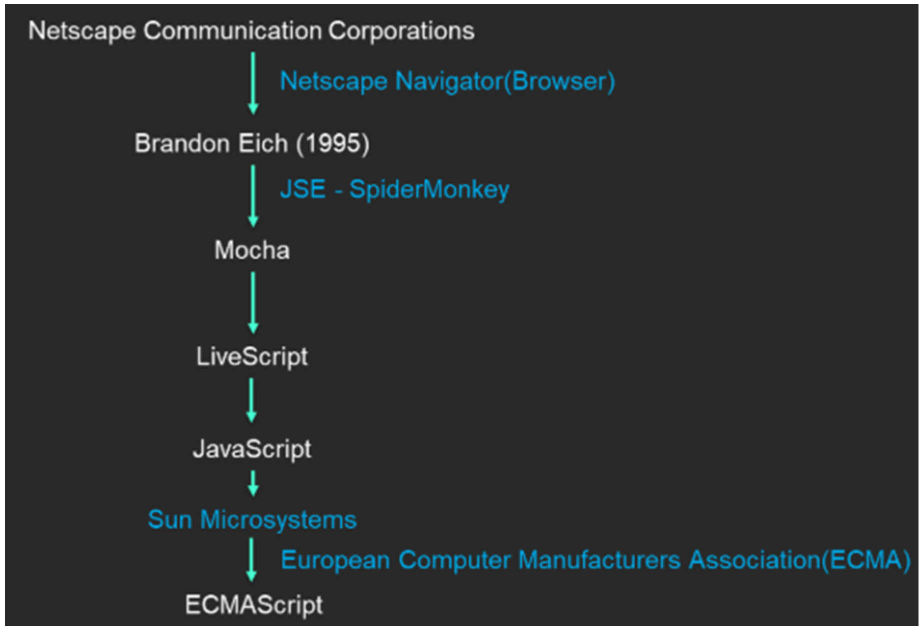
vi. Hiring *Brendan Eich* to embed the Scheme language.

vii. They collaborated with Brendan Eich to create JavaScript.

viii. Early *1995* ⇒ The scripting language was initially called "*Mocha*".

ix. September 1995 ⇒ Renamed "*LiveScript*"

x. December 1995 ⇒ Final name, "*JavaScript*" , was used in Netscape Navigator 2.0



3. ECMA:

a. ECMA stands for (*European Computer Manufacturers Association*).
b. ECMA International, is an organization that develops and publishes technical standards.
c. Its role in the development of JavaScript is critical due to the need for standardization across different web browsers.
d. Since 2015, major versions have been published every *June*.
e. ECMAScript 2025 (*ES2025*), the 16th and current version, was released on *25th, June 2025*.

4. Characteristics of JS:

a. Client-Side Scripting Language

i. Runs in the browser to make web pages interactive.
ii. Example: Show a popup when a button is clicked.

b. **Dynamically Typed Programming Language**

   i. You don't need to declare the type of variables (string, number, etc.).
   ii. JS Automatically detects the type at runtime.
   iii. let x = 5; // number
   iv. x = "Hello"; // now string

c. **Weakly Typed Programming Language**

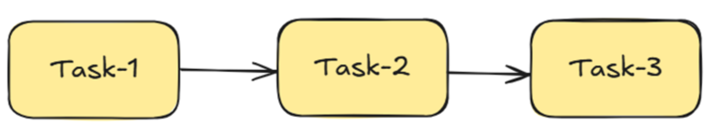
   i. Allows operations between different data types.
   ii. JS automatically converts types if needed.
   iii. Example:
   iv. console.log(5 + "2"); // "52" (number + string = string)

d. **Interpreted Language**

   i. Runs directly without compiling.
   ii. You can just write code and execute it.

e. **Single Threaded**

   i. JavaScript runs one task at a time.
   ii. Think of it as one person doing tasks in order.



f. **Asynchronous in Nature**
    i. Can do other work while waiting for something (like a server response).
    ii. Example: Fetch data from a server without freezing the page.

g. **Prototype-Based Object-Oriented**
    i. Objects in JS **inherit** features from other objects using prototypes, not classes (though ES6 has class syntax).

h. **Run in Browsers and Servers**
    i. Browser → Frontend (client-side)
    ii. Node.js → Backend (server-side)

i. **Cross-platform**
    i. Works on Windows, Mac, Linux, and mobile devices.

j. **Client Side Validation**
    i. Can check form data before sending it to the server.
    ii. Example: Show an error if a password is too short.

5. Difference Between Programming Language vs Scripting Language

Sr.No.	Programming Language	Scripting Language
1.	A programming language is a computer language that is used to communicate with computers using a set of instructions.	A scripting language is a type of programming language designed for a runtime system to automate the execution of tasks.
2.	It is compiled language or compiler-based language.	It is interpreted language or interpreter based language
3.	It is used to develop an application or software from scratch.	It is used to combine existing components and automate a specific task
4.	It runs or executes independently and does not depend on the parent (exterior) program.	It runs or executes inside another program.
5.	All programming languages are not scripting languages.	All scripting languages are programming languages.
6.	Usually, programming languages do not support or provide very little support for user interface designing, data types, and graphic designing.	Scripting languages provide great support to user interface design, data types, and graphic design.
7.	Examples → C, C++, Java, Scala, COBOL, etc.	Examples → Perl, Python, JavaScript, etc.